

JANUARY 2024

MANUFACTURING PROCESSES - I*Time Allowed: 2.5 Hours**Full Marks: 60*

Answer to Question No. 1 of 'Group: A' must be written in the main answer script. In Question No. 1, out of 2 marks for each MCQ, 1 mark is allotted for right answer and 1 mark is allotted for correct explanation of the answer.

Answer any Five (05) Questions from 'Group: B'.

GROUP: A

1. Choose the correct answer from the given alternatives and explain your answer (any ten): **2x10=20**

- i. The metal is subjected to mechanical working for –
a) Refining grain size b) Changing original block into desired shape c) Controlling the direction of flow lines d) All of these
- ii. The hot working of metals is carried out –
a) At the recrystallization temperature b) Below the recrystallization temperature c) Above the recrystallization temperature d) At any temperature
- iii. Structural sections such as rails, angles, I-beams are made by –
a) Hot rolling b) Hot drawing c) Hot piercing d) Hot extrusion
- iv. In indirect extrusion, which type ram is used?
a) Solid ram b) Hollow ram c) Curved ram d) None of these
- v. In a four high rolling mill, there are four rolls out of which –
a) One is working roll and three are backing up rolls b) Two are working rolls and two are backing up rolls c) Three are working rolls and one is backing up roll d) All of the four are working rolls
- vi. In oblique cutting of metals, the cutting edge of the tool is –
a) Perpendicular to the work piece b) Perpendicular to the direction of tool travel c) Parallel to the direction of tool travel d) Inclined at an angle less than 90° to the direction of tool travel
- vii. Among the following, which one is considered as production purpose Lathe?
a) Center Lathe b) Gap bed Lathe c) Engine Lathe d) Capstan and Turret Lathe
- viii. Which of the following is related with offsetting the tail stock?
a) Turning b) Facing c) Taper turning d) Knurling
- ix. In order to hold an irregular cross-sectional job in a Lathe, which type of chuck is suggested?
a) Three jaw self-centered chuck b) Four jaw independent chuck c) Collet chuck d) Magnetic chuck
- x. The time elapse between two successive grinding of a cutting tool is known as –
a) Machining time b) Tool life c) Set-up time d) None of these
- xi. The main functions of cutting fluids are –
a) Lubrication b) Cooling c) Lubrication and cooling d) None of these
- xii. Which form of metal is used in casting process?
a) Solid b) Liquid c) Gas d) Plastic
- xiii. In which casting process master pattern and wax pattern are used?
a) Continuous casting b) Centrifugal casting c) Investment casting d) Shell casting
- xiv. Among the following, which one is fall under the category of pressure welding?
a) TIG welding b) Arc welding c) Resistance welding d) None of these
- xv. Among the following, which welding process does not require any electric power supply during its execution?
a) TIG welding b) MIG welding c) Thermit welding d) Resistance welding

GROUP: B

2. a) State the difference between hot working and cold working processes.
b) State the advantages and disadvantages of forging process.
3. a) With a suitable diagram explain the working of Four High Rolling Mill.
b) Describe with neat sketch the working principle of Direct Extrusion Process.

[4+4]

[4+4]

4. a) Define Tool Life of a cutting tool. How tool life of a cutting tool can be measured?
b) Explain the given tool signature of a single point cutting tool: 10° , 8° , 7° , 6° , 15° , 30° , $1/8$ (inch)

[3+5]

5. a) Explain the forces acting during oblique cutting operation by a single point cutting tool.
b) What are the desirable properties of cutting fluid?

[3+5]

6. a) What is taper? How taper can be represented?

b) A shaft 600 mm long is to be turned taper for a length of 225 mm. The amount of taper is 1:100. If the shaft diameter is 75 mm, determine the smaller diameter, half taper angle and set-over required to turn the taper.

[3+5]

7. With suitable diagram explain the process of investment casting.

[8]

8. a) State about the defects which may generate in casting process.

b) With suitable diagram explain the working principle of arc welding.

[3+5]

9. a) Explain the different types of welding flames used in gas welding.

b) State the difference between MIG welding and TIG welding.

[4+4]

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